

YUBRAJ SHAHI

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate (Pokhara University, 2025) with hands-on expertise in embedded systems, IoT, and robotics. Led hardware-software integration on a real-world IoT safety device (Smart Helmet) and completed a year-long R&D internship at the Robotics Association of Nepal. Currently delivering STEAM education and mentoring students in robotics and programming. Holds foundational training in cloud computing (AWS), cybersecurity, and generative AI. Actively pursuing opportunities in Networking, Cybersecurity, Cloud, or IoT — at entry-level, internship, or freelance level.

EDUCATION

Bachelor of Engineering — Electronics & Communication Engineering

2021 – 2025

Cosmos College of Management and Technology, Pokhara University • Kathmandu, Nepal

Full Scholarship Recipient • *Relevant Coursework:* Computer Networks, Cybersecurity, Cloud Computing, Database Management Systems, Operating Systems, Telecommunications, Wireless & Digital/Analog Communication, C#

TECHNICAL SKILLS

Programming Languages: Python (Intermediate), C (Intermediate), C++ (Intermediate), Embedded C/C++ via Arduino IDE (Advanced)

Microcontrollers & Embedded: Arduino Uno (Advanced), ESP32 (Advanced), ESP-NOW Protocol, PCB Design (EasyEDA), Proteus Simulation

Networking & Security: CCNA Fundamentals, VLANs, Routing & Switching, Basic Linux CLI

Cloud & DevTools: AWS (Cloud Practitioner level), VS Code

Web & Design: HTML, CSS, Canva, SEO, Content Writing

Productivity: MS Office Suite, Google Workspace Essentials

WORK EXPERIENCE

STEAM Teacher | Nepatronix

Jan 2026 – Present

- Delivered hands-on STEAM lessons covering robotics, electronics, and programming to school-level students.
- Designed and facilitated interactive, project-based workshops using Arduino and sensors.
- Guided students in building mini-projects, fostering problem-solving skills and technical creativity.
- Supported curriculum development and classroom management for technical training programmes.

R&D Intern | Robotics Association of Nepal (RAN)

Nov 2023 – Oct 2024

- Designed and prototyped Arduino-based robotic systems with circuit simulation using Proteus and EasyEDA.
- Integrated sensors and microcontrollers for embedded system applications, improving hardware-software interaction.
- Mentored school-level students in robotics concepts and hands-on project development.
- Collaborated on research-oriented and educational robotics projects as part of a cross-functional team.

LEADERSHIP & VOLUNTEER EXPERIENCE

Advisor | Hult Prize — Cosmos College 2025 – 2026

- Mentored student teams on idea validation, business model development, and pitch preparation.
- Supported startups in refining concepts, improving presentations, and executing project plans.

Deputy Campus Director | Hult Prize — Cosmos College 2024 – 2025

- Led and coordinated entrepreneurship events, workshops, and competitions across campus.
- Managed student teams, logistics, and programme execution for all Hult Prize activities.

Graphics Designer | Hult Prize — Cosmos College 2023 – 2024

- Designed promotional posters and social media content, maintaining consistent visual branding.

Event Organiser | Robotics for Good Youth Challenge & Young Scientists Conference 2025

- Managed event operations, participant coordination, and technical session logistics for both national-level events.

Mentor | Python Workshop — Girls in ICT Nepal 2025

- Taught Python programming fundamentals and guided participants through hands-on coding exercises.

KEY PROJECT

Smart Helmet for Accident Prevention and Detection | Final Year Project

Role: Embedded System Developer & Team Member • Technologies: ESP32, Arduino IDE (C/C++), ESP-NOW, GPS (NEO-10M), GSM (A7670C), MPU6050, MQ-3, IR Sensor, EasyEDA PCB Design

- Developed an IoT-based safety helmet that detects accidents using accelerometer/gyroscope (MPU6050) and sends real-time GPS location via GSM SMS alert.
- Implemented alcohol detection (MQ-3 sensor) to block ignition and helmet-wear detection (IR sensor) to enforce safety compliance.
- Engineered wireless communication between helmet and bike units using the ESP-NOW protocol, eliminating internet dependency.
- Delivered a working prototype demonstrating cost-effective road safety improvement for real-world motorcycle conditions in Nepal.

CERTIFICATIONS & TRAINING

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- **AWS Academy Cloud Architecting [149483]** — AWS Academy, 2025 *[In Progress]*
 - **AWS Certified Cloud Practitioner Preparation** — AWS / Online Platform, 2025
 - **Generative AI Mastermind** — Outskill (Online), 2025
 - **Python Programming (as Mentor)** — Girls in ICT Nepal / Cosmos College, 2025
 - **Google Tools Essentials** — Nobel Internship Program, 2025
 - **Biomedical Instrumentation Engineering** — Pokhara University (Academic), 2024

KEY ACHIEVEMENTS

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- Awarded Full Scholarship for B.E. in Electronics & Communication Engineering (Pokhara University).
 - 1st Runner-Up — EEPEX Circuit Design Competition, Kathmandu University.
 - Successfully built and demonstrated the Smart Helmet prototype as final year capstone project.
 - Organised the Robotics for Good Youth Challenge and Young Scientists Conference 2025 (national level).
 - Participated in the South Asia Drone Forum 2025, gaining exposure to emerging drone technologies.
 - Progressed through three leadership roles at Hult Prize (Graphics Designer → Deputy Campus Director → Advisor).

LANGUAGES

Nepali — Native **English** — Professional Working Proficiency **Hindi** — Conversational